

Partial Result: A Positive-Area Support Criterion for the Harmonic Bergman Question

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Status

This is a partial positive answer to Question 2 in Han–Qiu–Wang, *Complete weighted Bergman spaces have bounded point evaluations*, arXiv:2111.07575. The source asks whether completeness of a weighted harmonic Bergman space $B^p(\Omega, \mu)$ implies locally uniformly bounded point evaluations. The source proves this when $\text{int supp } \mu$ is nonempty. The result below replaces that assumption by the weaker condition that $\text{supp } \mu$ has positive planar Lebesgue measure.

Theorem

Theorem 1. *Let $\Omega \subset \mathbb{C}$ be a domain, let μ be a positive Radon measure on Ω , and let $1 \leq p < \infty$. Define*

$$B^p(\Omega, \mu) = \left\{ f \in \text{Harm}(\Omega) : \int_{\Omega} |f|^p d\mu < \infty \right\}$$

with its natural $L^p(\mu)$ norm. Suppose that $B^p(\Omega, \mu)$ is a Banach space and that $\text{supp } \mu$ has positive two-dimensional Lebesgue measure. Then $B^p(\Omega, \mu)$ has locally uniformly bounded point evaluations: for every compact set $K \subset \Omega$ there is a constant C_K such that

$$\sup_{z \in K} |f(z)| \leq C_K \left(\int_{\Omega} |f|^p d\mu \right)^{1/p} \quad (f \in B^p(\Omega, \mu)).$$

Proof. Choose a compact set $E \subset \text{supp } \mu$ with positive planar Lebesgue measure. This is possible by inner regularity of planar Lebesgue measure, using that Ω is sigma-compact.

Fix a compact set $K \subset \Omega$. Since Ω is a domain, there is a relatively compact subdomain $S \Subset \Omega$ such that $K \cup E \subset S$. Let $h^\infty(S)$ be the Banach space of bounded harmonic functions on S with the supremum norm, and consider the restriction map

$$R_S : B^p(\Omega, \mu) \longrightarrow h^\infty(S), \quad R_S f = f|_S.$$

This map is well-defined because every harmonic function on Ω is bounded on relatively compact subdomains.

We show that R_S has closed graph. Let $f_n \rightarrow f$ in $B^p(\Omega, \mu)$ and $f_n|_S \rightarrow g$ in $h^\infty(S)$. Since $\overline{S}$ is compact and μ is Radon, $\mu(S) < \infty$. Therefore $f_n|_S \rightarrow f|_S$ in $L^p(S, \mu)$ and also $f_n|_S \rightarrow g$ in $L^p(S, \mu)$. Hence $f|_S - g = 0$ μ -almost everywhere on S .

The function $f|_S - g$ is harmonic on S . By continuity and the definition of support, a continuous function that vanishes μ -almost everywhere on S vanishes on $\text{supp } \mu \cap S$. In particular, $f|_S - g$

vanishes on E . Since E has positive planar measure and a nonzero harmonic function has a zero set of planar measure zero, it follows that $f|_S - g \equiv 0$ on S . Thus the graph of R_S is closed.

The closed graph theorem gives a constant C_S such that

$$\|f\|_{h^\infty(S)} \leq C_S \|f\|_{B^p(\Omega, \mu)}.$$

Since $K \subset S$, this implies the desired estimate with $C_K = C_S$. □

Comment

The condition $|\operatorname{supp} \mu| > 0$ is strictly weaker than $\operatorname{int} \operatorname{supp} \mu \neq \emptyset$: a fat Cantor set of positive area inside Ω has empty interior but is still a uniqueness set for harmonic functions. Thus the theorem expands the source's stated partial result for the harmonic question, but it does not settle the remaining zero-area uniqueness-set cases.

References

- Y. Han, Y. Qiu, and Z. Wang, *Complete weighted Bergman spaces have bounded point evaluations*, arXiv:2111.07575.